Daleel 2026 Shared Task on Arabic Argumentative Discourse Mining

Sara Nabhani
University of Groningen
s.nabhani@rug.nl

Nahla Bassyouni
QatarDebate Center
nhb15@mail.aub.edu

Ali Al-Zawqari
Vrije Universiteit Brussel
aalzawqa@vub.be

Mohammad Khader
QatarDebate Center
mkhader@qatardebate.org

Khalid Al-Khatib
University of Groningen
khalid.alkhatib@rug.nl

Abstract

This paper presents Daleel 2026, the first shared task on argument mining in Arabic news editorials and competitive debates. Daleel includes two tasks: Task 1 identifies which of six argumentative discourse unit (ADU) categories are present in a paragraph, while Task 2 locates the units with their labels and exact text spans. The tasks are performed using a new corpus of 30 documents, 1,042 paragraphs, and about 76k words. Twenty-nine teams registered, 12 participated, and 10 submitted system papers. Both tasks have Closed and Open tracks. The best Closed-track results were 71.17 macro-F1 points for Task 1 and 75.47 modified-overlap F1 points for Task 2. The results show different insights: the best Closed-track score of each team was higher on debates than on editorials. A broad range of approaches was used in Task 1 submissions, including Arabic encoders, large language models, rules, and ensembles. Task 2 systems, in particular the top-ranked ones, were more similar and mainly used CRF or Viterbi decoding with combinations of models or training runs. The rare ADU categories remained difficult in both tasks, and identifying exact span boundaries was an additional challenge in Task 2.

1 Introduction

Argument mining aims to identify and analyze the argumentative structure of natural language text. It supports the automatic study of how arguments are expressed and how different parts of an argument function within a text (Lawrence and Reed, 2019). A central element in this process is the identification of argumentative discourse units (ADUs), which represent the text spans that serve an argumentative elements. Argument mining therefore involves both locating these units and determining the role they play in the argument (Stab and Gurevych, 2017). These tasks have been studied across several domains and annotation schemes. However, most established resources and methods have focused on English.

In recent years, argument mining in Arabic has received growing attention, but annotated resources remain limited. Recent work has introduced resources for Arabic competitive debates, including Munazarat, and for student argumentative writing through the Qatari Corpus of Argumentative Writing (QCAW) (Khader et al., 2024; Zaghouani et al., 2024). Other work has studied the classification of argument elements and styles in Arabic debates (Al-Zawqari et al., 2025). These resources provide important foundations, but they address different argumentation schemes and tasks. More recently, (Nabhani and Al Khatib, 2025) studied span-level argument mining in Arabic without native annotations, using cross-lingual transfer and translated training data. This leaves a need for native Arabic annotations that support ADU identification and classification across different forms of argumentative discourse.

Daleel 2026 addresses this gap by introducing a shared task on Arabic argumentative discourse mining across: 1) news editorials and 2) competitive debates. The task uses six ADU categories: Common Ground, Assumption, Testimony, Statistics, Anecdote, and Other, following the argumentation strategy scheme introduced by (Al-Khatib et al., 2016). It defines two related tasks over the same paragraphs. Task 1 requires systems to identify the ADU categories present in a paragraph as a multi-label classification problem. Task 2 requires systems to locate the corresponding ADUs and identify their exact character spans and categories. Both tasks are offered under Closed and Open tracks, allowing the evaluation of systems under different resource settings.

This paper presents an overview of the Daleel 2026 shared task. We describe the task design, the annotation process, and the resulting Arabic corpus. We also present the evaluation setup, metrics, base-

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1287–1298
October 28-29, 2026 ©2026 Association for Computational Linguistics

lines, and the development and testing phases. We then summarize the approaches developed by the participating teams and report the official results across both tasks and tracks. Finally, we analyze the main patterns that emerge from the shared task, with particular attention to genre differences, difficult ADU categories, span detection, system design choices, and cross-domain generalization.

2 Related Work

Argument mining research differs in how argumentative units and their roles are defined. Work on persuasive essays commonly distinguishes major claims, claims, premises, and the relations between them (Stab and Gurevych, 2014, 2017). Other work has addressed argumentative units in user-generated discourse (Habernal and Gurevych, 2017) or classified the semantic types of claims and premises (Hidey et al., 2017). A different view based on argumentation strategies in news editorials was proposed in (Al-Khatib et al., 2016), distinguishing six categories of evidence and discourse function. Later work examined how these strategies vary across topics (Al-Khatib et al., 2017).

Arabic argumentation resources have expanded in recent years and cover different forms of argumentative text. Munazarat 1.0 introduced a corpus of Arabic competitive debates (Khader et al., 2024). Subsequent work developed a hybrid annotation model for these debates and studied the neural classification of their argument elements and styles (Al-Sharafi et al., 2025; Al-Zawqari et al., 2025). Outside the debate setting, QCAW 1.0 provides a corpus of Arabic student argumentative writing (Zaghouani et al., 2024). Argumentation information has also been used as a feature in downstream Arabic tasks such as propaganda detection (Nabhani et al., 2025). These resources differ in genre, annotation scheme, and target task.

Modeling work on Arabic argument mining has also explored how to deal with the limited availability of native annotations. Cross-lingual transfer and translation-based approaches for Arabic argument mining without native annotations were studied in (Nabhani and Al Khatib, 2025), using an English argument mining corpus as the source data. In parallel, several pretrained language models have been developed for Arabic. AraBERT provides Arabic-specific Transformer representations (Antoun et al., 2020), and both ARBERT and MARBERT target Modern Standard Arabic and dialectal variation (Abdul-Mageed et al., 2021). CAMeLBERT further studies the effect of language variant, model size, and task type on Arabic language understanding (Inoue et al., 2021), while AraELECTRA provides an alternative discriminator-based pretraining approach (Antoun et al., 2021). Together, these models provide Arabic-specific backbones for classification and span detection tasks.

A separate line of research studies argument mining in debates and dialogue. In (Habernal and Gurevych, 2015), debate portals were used to study argument mining across domains in user-generated discourse. QT30 introduced a corpus of argument and conflict in broadcast debates (Hautli-Janisz et al., 2022), extending argument mining to spoken and transcribed interaction. Project Debater demonstrated argument mining and related technologies at the scale of an autonomous debating system (Slonim et al., 2021). More recently, DialAM-2024 introduced the first shared task on dialogical argument mining, combining propositional relations with speech illocutions (Ruiz-Dolz et al., 2024). This provides a relevant precedent for studying argumentation in interactive and spoken settings.

Shared tasks have also become an important part of Arabic NLP evaluation. AraIEval evaluated persuasion techniques and disinformation detection in Arabic, including span-level identification of persuasion techniques (Hasanain et al., 2023). More recently, the BAREC 2025 shared task evaluated Arabic readability assessment using multiple evaluation tracks (Elmadani et al., 2025). Such shared tasks provide common data, task definitions, and evaluation settings for comparing different modeling approaches. Daleel follows this shared-task setting for Arabic argument mining, combining paragraph-level ADU classification with span-level ADU detection. It also evaluates the two tasks across editorials and competitive debates, making genre and cross-domain performance part of the evaluation.

3 Shared Task Description

3.1 ADU Taxonomy

Both tasks use the same taxonomy of six ADU categories. The taxonomy follows the argumentation strategy scheme introduced in (Al-Khatib et al., 2016). The categories describe how each unit functions within the argumentative discourse. Table 1 gives their definitions. An ADU may receive more

than one category when several definitions apply.

Table 2 gives four annotated examples from the corpus, with two examples from each genre and four different ADU categories.

3.2 Task 1: ADU Classification

Task 1 identifies the ADU categories present at the paragraph level. The input is an Arabic paragraph x from a news editorial or competitive debate. The label inventory is $\mathcal{L} = \{\mathrm{CO}, \mathrm{AS}, \mathrm{TE}, \mathrm{ST}, \mathrm{AN}, \mathrm{OT}\}$. A system predicts a subset $Y_1 \subseteq \mathcal{L}$. Since several categories may occur in the same paragraph, Task 1 is a multi-label classification problem.

3.3 Task 2: ADU Detection

Task 2 identifies the individual ADUs in a paragraph and assigns their categories. It uses the same paragraph input and label inventory as Task 1. A system predicts $Y_2 = \{(\ell_i, s_i, e_i)\}_{i=1}^{m}$. Each triple gives an ADU category $\ell_i \in \mathcal{L}$ and its start and end character offsets s_i and e_i. The interval $[s_i, e_i)$ marks the corresponding span in the original paragraph. Multiple triples may share a span when an ADU has more than one category. Task 2 therefore combines span detection with ADU classification.

3.4 Relationship between the Tasks

Both tasks operate on the same paragraphs and use the same label inventory. Task 1 records only whether each ADU category occurs. Task 2 preserves the category and location of every unit. The Task 1 output can therefore be derived from the Task 2 output as

$$Y_1 = \{\ell_i \mid (\ell_i, s_i, e_i) \in Y_2\}. \quad (1)$$

This projection removes the span boundaries and repeated occurrences of the same category. The reverse operation is not possible because Task 1 does not contain boundary information. Figure 1 illustrates this relation.

3.5 Closed and Open Tracks

Both tasks are offered in Closed and Open tracks. The Closed track restricts the use of external data and larger or proprietary large language models (LLMs). The Open track relaxes these resource restrictions and allows additional resources. Both tracks permit prompting, fine-tuning, rule-based methods, and hybrid systems. Table 3 summarizes the main resource differences. Results for the two tracks are reported separately.

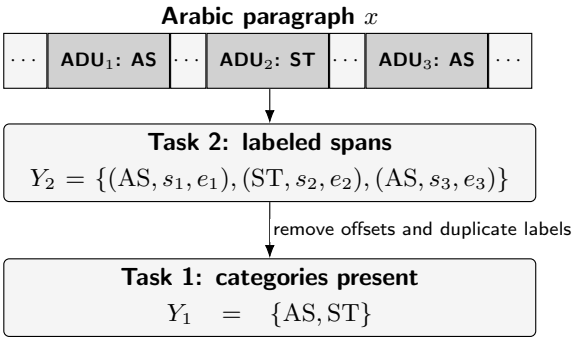


Figure 1: Relationship between the two tasks. Task 2 identifies each labeled span, and Task 1 retains only the distinct categories.

4 Dataset

Argumentation may vary depending on the type of text in which it occurs. Therefore, this shared task is designed not only to advance Arabic argument mining in general, but also to study how argumentation and argument mining differ across discourse genres. To investigate these differences, the shared task focuses on two genres: *Editorials* and *Debates*.

4.1 Data Sources

We constructed a corpus consisting of 20 news editorials and 10 debate transcripts, covering a range of topics and issues. Figure 2 shows the distribution of topics across editorials and debates in the corpus.

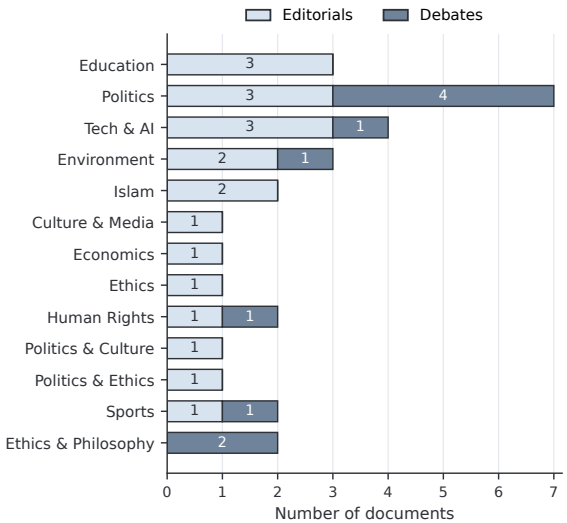


Figure 2: Topics distribution in the editorials and debate corpus.

Label	Category	Definition
AS	Assumption	States an assumption, conclusion, judgment, or opinion of the author, a general observation, possibly false fact, or similar. To make readers accept it, it is or would need to be supported by other units.
OT	Other	Does not, or hardly, add to the argumentative discourse, or does not match any of the other classes.
AN	Anecdote	Gives evidence by stating a personal experience of the author, an anecdote, a concrete example, an instance, a specific event, or similar.
TE	Testimony	Gives evidence by stating or quoting that a proposition was made by an expert, authority, witness, group, organization, or similar.
CO	Common Ground	States common knowledge, a self-evident fact, an accepted truth, or similar. It refers to general issues, not to specific events. Even if not known in advance, it will be accepted without proof or further support by all or nearly all possible readers.
ST	Statistics	Gives evidence by stating or quoting the results or conclusions of quantitative research, studies, empirical analyses of data, or similar. A reference may, but need not, be given.

Table 1: The six ADU categories used in Daleel 2026.

Genre	Category	Arabic example	English gloss
Editorial	Statistics (ST)	في عام 2024 ـ الذي شهد بالفعل تراجعات واضحة ـ حققت مرسيدس ربحا بلغ 10.4 مليار يورو	In 2024, despite clear declines, Mercedes made a profit of 10.4 billion euros.
Editorial	Testimony (TE)	تقول فيليز إن الناس لم يديروا ظهورهم للخصوصية، لكن كثيرين منهم يشعرون بالعجز حيالها	Velez says that people have not turned their backs on privacy, but many feel powerless about it.
Debate	Common Ground (CO)	حلف الناتو هو تحالف سياسي عسكري يقتضي أن يدافع عن الدول الموجودة فيه و الدول المشاركة فيه	NATO is a political and military alliance whose members are expected to defend the countries in the alliance.
Debate	Assumption (AS)	لابد من وضع سياسات و لا بد من التعاون الدولي بدون المساس بسيادة الدول لكي نعالج هذه المشكلة	Policies and international cooperation are needed to address this problem without infringing on state sovereignty.

Table 2: Annotated ADU examples from editorials and debates. The Arabic spans are taken directly from the corpus; English glosses are provided for readability.

Editorials. Editorials represent monological texts with extended argumentative writing, in which a single author develops and supports a position on an issue. The editorials were collected from different high-quality Arabic news portals (e.g., Al Jazeera, Al Araby, BBC, DW). Figure 3 shows the distribution of the selected editorials across these portals.

Debates. Debates represent dialogical argumentative discourse, in which multiple speakers present claims, respond to one another, and express agreement or disagreement. The debates were sampled from QatarDebate Center[1] tournament material.

4.2 Annotation process

The annotation process was based on the scheme proposed by Al-Khatib et al. (2016), originally developed to identify argumentative units in English texts. All annotators have a background in debating and participated in multiple training sessions, during which the scheme was slightly refined and adapted to suit the characteristics of Arabic Argumentation better. Each editorial and debate transcript was annotated independently by two annotators. Inter-annotator agreement (IAA) was measured using Soft γ (Mathet et al., 2015) and Intersection over Union (IoU), achieving average scores of 0.32 and 0.48, respectively.[2] Soft γ accounts for both label and positional mismatches, with disagreements in labels receiving a greater penalty than discrepancies in span position. The resulting annotations were then reviewed, and any disagreements were resolved by an expert in debate and

[1] https://qatardebate.org/

[2] IAA was calculated over 28 documents, excluding the two documents used for annotation training. IoU was calculated using a threshold of 0.5.

Resource	Closed	Open
Organizer training and development data	Yes	Yes
Public pretrained models and standard NLP tools	Yes	Yes
Open-weight LLMs ($\leq$70B parameters)	Yes	Yes
Additional datasets or external resources	No	Yes
Proprietary LLMs	No	Yes
Open-weight LLMs (>70B parameters)	No	Yes
External retrieval, data augmentation, or translation	No	Yes

Table 3: Allowed resources in the Closed and Open tracks.

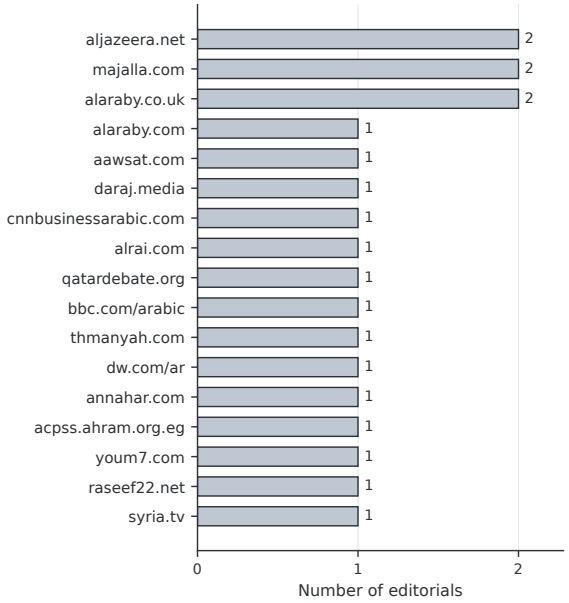


Figure 3: Number of editorials per news portal in the editorials corpus.

argumentation. During adjudication, the annotations produced by both annotators were compared against the guidelines, and, in cases of disagreement, the annotation that most closely adhered to the specified criteria was retained. Given the inherently subjective nature of label identification, some cases permitted multiple plausible interpretations. When both annotations could reasonably be supported by the guidelines, the decision was based on the interpretation most consistent with how similar cases had been handled across the dataset and by the annotators as a whole. Where neither initial annotation provided a sufficiently appropriate resolution, a new gold annotation was assigned based on the same interpretive rationale established in earlier adjudication decisions. The final annotation guidelines can be found in the shared task repository on GitHub.[3]

	Editorials	Debates	Overall
Corpus size			
Documents	20	10	30
Paragraphs	430	612	1,042
Words	22,733	53,327	76,060
Per-paragraph averages			
Words	52.87	87.14	72.99
Labels	1.57	1.92	1.78
ADUs	3.24	5.66	4.66

Table 4: Corpus statistics by genre.

4.3 Resulting Corpus

Table 4 shows summary statistics of the final corpus. Although the corpus is relatively small in terms of the number of documents, the documents are argument-rich. The 20 editorials average 1,137 words each, and the 10 debates average 5,333 words each, resulting in a total of approximately 76K words for the full corpus. The task is defined at the paragraph level. Therefore, the documents were segmented into paragraphs, resulting in a total of 1,042 paragraphs.

Figure 4 shows the distribution of ADU categories across the entire corpus, while Table 5 breaks this distribution down by genre.

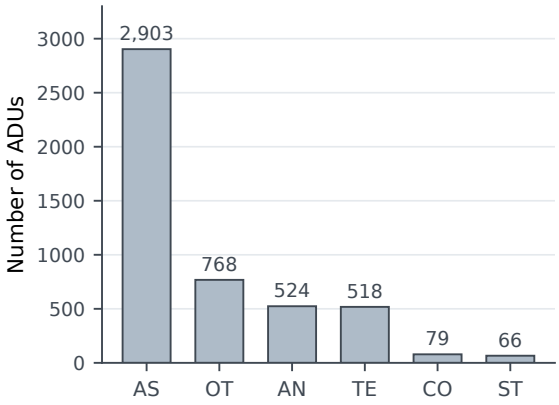


Figure 4: ADU category distribution across the corpus.

The final corpus is split into training, development, and test sets for the different stages of the shared task. The training set contains 612 paragraphs from 18 documents (6 debates and 12 editorials). The development set contains 217 paragraphs from 6 documents (2 debates and 4 editori-

[3] https://github.com/Argmining/Daleel2026

als), and the test set contains 213 paragraphs from 6 documents (2 debates and 4 editorials). This split ensures that all three sets include both genres and a comparable number of documents and paragraphs. The distribution of the ADU categories across the three data splits is shown in Table 5.

5 Evaluation

5.1 Development and Testing Phases

The shared task consisted of two phases: a development phase and an evaluation phase. In the development phase, participants were provided with the training data together with its gold annotations for both tasks, as well as the development input without annotations. Participants used these resources to develop and tune their systems and to evaluate their performance on the development set.

In the evaluation phase, which follows the development phase, participants received the gold labels for the development set and the input for the test set. Systems were then run on the test input, and the system outputs for the test set were used for the final evaluation and ranking.

In both phases, participants had the option of training their models using only editorials, only debate, or both genres. All submitted systems were subsequently evaluated on three test settings: editorials only, debate only, and the combination of both.

5.2 Metrics

Task 1 - ADU Classification. We use Macro F1 over the classes.

Task 2 - ADU Detection. We use an F1 measure adapted to account for partial matching between predicted and gold spans, following the evaluation measure used in previous shared tasks (Hasanain et al., 2024; Alam et al., 2022; Dimitrov et al., 2021). A predicted span is considered correct if it overlaps with a gold span and has the correct label, with the score adjusted according to the degree of overlap.

Let s be a predicted span, t a gold span, $l(s)$ and $l(t)$ their respective labels, and $|s \cap t|$ the length of the character overlap between s and t. The credit recevied by the (s, t) pair is defined as:

$$C(s, t, h) = \frac{|s \cap t|}{h} \quad \text{if } l(s) = l(t),$$

where h is the normalization length. For precision, we use the length of the predicted span:

$$P = \frac{1}{|S|} \sum_{s \in S} \sum_{t \in T} C(s, t, |s|).$$

For recall, we use the length of the gold span:

$$R = \frac{1}{|T|} \sum_{s \in S} \sum_{t \in T} C(s, t, |t|).$$

The final score is the harmonic mean of precision and recall:

$$F_1 = \frac{2PR}{P + R}.$$

5.3 Baselines

We provide a random baseline for both tasks.

Task 1 - ADU Classification. For Task 1, the baseline first samples a number of labels between 1 and 4 for each paragraph. It then assigns ADU categories randomly, taking their distribution in the corpus into account.

Task 2 - ADU Detection. For Task 2, the baseline similarly samples a number of ADU spans between 1 and 4 per paragraph. For each ADU span, it selects the categories at random, taking their distribution in the corpus into account. The start offset of each ADU span is chosen randomly between the beginning and the end of the paragraph, and the end offset is chosen randomly between the start offset and the end of the paragraph.

6 Participants and Systems

A total of 29 teams registered for the shared task, 12 teams participated, and 10 teams submitted system description papers. Five teams covered both tasks, three covered only Task 1, and two covered only Task 2. *TTLab* (Verma et al., 2026) submitted a ranked Task 1 run, although its paper describes only Task 2. Of the teams with papers, only *Durham-Jazan* (Saad Alshehri et al., 2026) entered both the Closed and Open tracks. *REGLAT* also has a ranked Task 1 Open run, but no system paper was available. Table 6 lists each team's final run in every task–track combination it entered.

6.1 Task 1 Systems

Closed track. Arabic encoders appeared in most Task 1 systems, either alone or as part of a hybrid. *Northwestern_12* (Sedqy Shlkamy and Zaghouani, 2026) fine-tuned CAMeLBERT-Mix with class-weighted loss and a separate threshold for each label. *BURHAN* (Osrf et al., 2026) fine-tuned

Label	Genre		Data split (both genres)		
	Editorials	Debates	Train	Dev.	Test
AS	793 (27.3%)	2,110 (72.7%)	1,739 (59.9%)	587 (20.2%)	577 (19.9%)
OT	42 (5.5%)	726 (94.5%)	492 (64.1%)	126 (16.4%)	150 (19.5%)
AN	289 (55.2%)	235 (44.8%)	343 (65.5%)	111 (21.2%)	70 (13.4%)
TE	181 (34.9%)	337 (65.1%)	313 (60.4%)	112 (21.6%)	93 (18.0%)
CO	36 (45.6%)	43 (54.4%)	50 (63.3%)	16 (20.3%)	13 (16.5%)
ST	52 (78.8%)	14 (21.2%)	38 (57.6%)	9 (13.6%)	19 (28.8%)

Table 5: Distribution of ADU categories by genre and data split. Genre shows the distribution between editorials and debates; data split shows the distribution across training, development, and test with both genres combined.

Team	Affiliation	Task 1		Task 2	
		Closed	Open	Closed	Open
BME-Daleel	Budapest University of Technology and Economics	2 (71.17)	–	1 (75.47)	–
Durham-Jazan	Durham University; Jazan University; University of Waikato	5 (65.68)	1 (66.93)	6 (72.92)	1 (73.61)
NU_Analytics	Precision AI; Nile University; Helwan University	4 (67.14)	–	2 (75.23)	–
SalahAbdo	Independent researcher	1 (71.17)	–	5 (73.16)	–
TTLab	Text Technology Lab, Goethe University Frankfurt	6 (64.51)	–	3 (73.74)	–
AIDAL-ARG	IRIT, Université Toulouse Capitole	7 (59.92)	–	8 (68.32)	–
RabeeQasem–ImpactAi	GGateway, Nablus, Palestine	–	–	4 (73.23)	–
Northwestern_12	Northwestern University in Qatar	8 (56.48)	–	–	–
BURHAN	University College of Applied Sciences; King Fahd University of Petroleum & Minerals	9 (55.97)	–	–	–
Solodiscourse	Independent researcher	10 (52.65)	–	–	–
OliveTrees†	–	3 (68.82)	–	7 (72.36)	–
REGLAT†	–	–	2 (59.13)	–	–
Baseline	–	11 (36.44)	3 (36.44)	9 (28.53)	2 (28.53)

Table 6: Daleel 2026 participating teams and the organizer baseline. Each result cell gives the official rank followed by the score in parentheses. Task 1 uses macro-F1; Task 2 uses modified overlap F1. Scores are reported on a 0–100 scale and rounded to two decimal places. A dash means that the team had no final ranked submission in that task and track. † No accompanying system paper was available.

MARBERTv2 with label-weighted focal loss. It used multilabel-stratified five-fold cross-validation, calibrated a threshold for each label, ensembled the folds, and added a lexical rule for Statistics. *Solodiscourse* (Hussain Usmani, 2026) added handcrafted lexical features to CAMeLBERT-Mix and combined focal loss with a five-fold ensemble.

LLMs played a larger role in three submissions. *SalahAbdo* (Ali Abdo, 2026) assigned different labels to different components: Gemma voting, a CAMeLBERT classifier, signals derived from Task 2, or a fixed prediction-count rule for Common Ground. An LLM also verified Statistics predictions. *NU_Analytics* (Ahmad Saeed Abdelaziz et al., 2026) compared joint label generation with Qwen3.5-9B and independent yes/no scoring with ALLaM-7B; its final submission used ALLaM-7B. *Durham-Jazan* (Saad Alshehri et al., 2026) split the task by genre, using Qwen3-32B for editorials and an encoder ensemble for debates. *BME-Daleel* (Mohaisen, 2026) fused probabilities from paragraph-level, lexical, and CRF-based components. *AIDAL-ARG* (Salas-Jimenez et al., 2026) compared multitask CAMeLBERT models with a two-stage BIO segmentation and classification pipeline; an LLM settled ambiguous label decisions.

Open track. *Durham-Jazan* (Saad Alshehri et al., 2026) added 291 synthetic paragraphs and fused its paragraph classifier with labels derived from Task 2 span predictions.

6.2 Task 2 Systems

Closed track. Most Task 2 systems used sequence labeling, often with a structured decoder. *BME-Daleel* (Mohaisen, 2026) combined several en-

coder and CRF sources, projected their token distributions onto a common grid, and applied constrained Viterbi decoding. *NU_Analytics* (Ahmad Saeed Abdelaziz et al., 2026) adapted AraBERT-large to the task, trained BIO-CRF models with boundary-jitter augmentation, and combined four seeds in character space. *TTLab* (Verma et al., 2026) placed a two-layer BiLSTM and constrained CRF above MARBERTv2. It averaged emissions from five folds before a single Viterbi decode. *RabeeQasem–ImpactAi* (Al-Qasem, 2026) used an AraBERT encoder with a Conformer tagging head, processed long paragraphs in overlapping windows, and post-processed the resulting spans. *Durham-Jazan* (Saad Alshehri et al., 2026) used separate genre branches: a CAMeLBERT ensemble for editorials and a MARBERTv2 ensemble for debates, followed by metric-aware span compaction. *SalahAbdo* (Ali Abdo, 2026) first asked an LLM for label–quotation pairs. The quotations were aligned with the paragraph, after which a classifier and Viterbi decoding revised their labels. *AIDAL-ARG*'s best official run was a genre-specific multi-task ensemble. A separate two-stage pipeline used an LLM only when label decisions were uncertain or inconsistent.

Open track. *Durham-Jazan* (Saad Alshehri et al., 2026) added 158 synthetic span paragraphs and blended character-level probabilities from its CAMeLBERT and MARBERTv2 branches.

7 Results and Discussion

The best-ranked run from every team outperformed the corresponding baseline. In Task 1 Closed, *SalahAbdo* and *BME-Daleel* had nearly identical scores: both round to 71.17, with *SalahAbdo* ranked first. Task 2 Closed was also close at the top; 2.55 separated first and sixth place.

Only *Durham-Jazan* has documented submissions in both tracks. *Durham-Jazan*'s Open runs scored 1.25 higher in Task 1 and 0.69 higher in Task 2 after the team added synthetic data and changed how the component outputs were combined. These changes were made together, so the gains cannot be attributed to either one. No other team provides a non-identical Closed–Open comparison.

7.1 Genre and Cross-Domain Effects

Every team scored better on debates than on editorials in its best Closed-track run. Averaged across the ten Task 1 and eight Task 2 runs, with each team counted once, the debate-minus-editorial difference was 18.8 and 12, respectively. The training split contains 357 debate paragraphs and 255 editorial paragraphs, and the label frequencies also differ between the two genres. We therefore cannot tell from these results alone whether the difference is due to genre, training size, label balance, or a combination of these factors.

Per-genre scores were not reported in all system papers, but the available results show the same pattern. In Task 1, *BME-Daleel* reported 62.39 on editorials and 77.78 on debates, *Durham-Jazan* 50.10 and 74.10 in the Closed track, and *NU_Analytics* 47.51 and 74.18. In Task 2, *BME-Daleel* reported 66.53 on editorials and 79.41 on debates, *Durham-Jazan* 61.00 and 76.20, *NU_Analytics* 64.17 and 79.31, and *TTLab* 63.56 and 77.84. These numbers cover only the teams that reported comparable per-genre results.

Some teams also submitted runs trained on both genres and runs trained on one genre only. Whenever both were available, the mixed-genre run had the highest overall score. *TTLab* showed this in both tasks. Its Task 1 score was 64.51 with mixed-genre training, 60.19 with debate training, and 54.65 with editorial training. The corresponding Task 2 scores were 73.74, 70.68, and 61.25. *Durham-Jazan* and *REGLAT* followed the same ordering in Task 1, as did *Solodiscourse* across its two submitted settings. These were not controlled comparisons, so we cannot attribute the higher scores to mixed-genre training alone.

7.2 Label and Span-Level Challenges

Common Ground was the one Task 1 label that several papers identified as especially difficult. *NU_Analytics* called it its hardest label. *AIDAL-ARG* reported zero F1 for all three variants in its development and test results, while *BURHAN* obtained 9.5 F1 on the official test set. *SalahAbdo* used a fixed prediction-count rule after its other detectors remained unreliable.

The difficulty appeared in encoder, LLM, and hybrid systems, so it is not confined to one model family. Its rarity and semantic overlap with Assumption are two possible reasons. Statistics was another label that teams handled separately, using lexical rules, targeted augmentation, oversampling, or LLM verification.

The available Task 2 analyses point to boundary errors, not only label errors. *BME-Daleel* exam-

ined 961 gold spans in the development set. Only 15.9% matched exactly; another 59.0% had the correct label but imperfect boundaries. *AIDAL-ARG* reported exact boundaries for 25.8–27.7% of its matched development spans. *NU_Analytics* found that missed spans were markedly shorter than matched spans. It also noted that about 1% of paragraphs contain overlapping gold spans, which a single BIO sequence cannot represent.

7.3 System Design Trends

Task 1 did not converge on one architecture. Some teams used encoder classifiers with weighted losses and calibrated thresholds. Others assigned labels to different LLM or rule-based components, or fused signals from paragraph, token, lexical, and cross-task models. Genre-aware routing and ensembling appeared in several systems.

The three highest-ranked documented Task 2 systems in the Closed track all used CRF or Viterbi decoding and combined multiple folds, seeds, or models. None of the documented LLM-based systems prompted the LLM to return free-form character offsets. *SalahAbdo* aligned generated quotations with the source text, *AIDAL-ARG* used the LLM only to settle label decisions, and *Durham-Jazan* reconstructed ordered labeled segments and obtained offsets from their concatenation. In these systems, the LLM supplied text or labels; alignment or structured decoding fixed the boundaries.

7.4 Generalization and Future Evaluation

More editorial training examples, especially for rare labels, would help test whether the gap comes from genre, sample size, or label balance. The current runs cannot separate these explanations.

Individual components are also hard to compare across papers because teams used different development splits, cross-validation setups, and model-selection procedures. *BURHAN*, for example, reported about 68 macro-F1 in internal cross-validation but 56 on the official test set. A shared held-out analysis split would make these comparisons more reliable.

For Task 2, evaluation should separate label and boundary errors. A single BIO sequence cannot represent overlapping spans, and *NU_Analytics* found that short spans were more likely to be missed. Future work should compare representations that allow overlapping spans with BIO tagging and report performance by span length. This would show whether a system chose the wrong label, the wrong boundary, or both.

8 Conclusion

Daleel 2026 provides the first benchmark for Arabic argument mining in news editorials and competitive debates. Twenty-nine teams registered, 12 took part, and 10 teams submitted system description papers. The submitted systems used different methods, and all performed better than the baseline. The results give initial reference scores for identifying ADU categories and locating their spans in Arabic text. We plan to extend the corpus with political speeches and annotate the argumentative relations between ADUs. The relation annotations will add another level to the current tasks by showing how the identified units are connected. The extended corpus may later support new Daleel shared tasks on relation identification and argument mining across genres.

Limitations

The Daleel corpus comprises 30 documents and 1,042 paragraphs. Although the documents contain many argumentative units, the dataset remains relatively small. The corpus also covers only news editorials and competitive debates, so the results may not generalize to other forms of argumentative discourse. Also, the texts are limited to Modern Standard Arabic and do not cover Arabic dialects or code-switched language. Some ADU categories have few examples. Common Ground was particularly difficult for several systems. The results for these categories should therefore be interpreted carefully and may change as more annotated data become available.

The debate data come from competitive tournaments, so the results may not generalize to other forms of spoken or dialogical argumentation. Some annotation decisions were also difficult, especially when deciding the exact span or label. Although these cases were adjudicated, the agreement scores show that the annotation is not always clear-cut.

The two genres also differ in the number of training paragraphs and in their label distributions, so the higher scores on debates cannot be attributed to genre alone. In Task 2, the overlap-based F1 combines span detection and labeling in one score, which makes it difficult to tell whether an error comes from the label, the span boundary, or both.

Acknowledgments

The organizers thank Ammar Al Sharaa, Hamzah Zaybak, Mohamed Albuainain, Nidal El Ibrik, Omar Akoum, and Rawan Herchi for their contribution to the annotation process. We also thank Roxanne El Baff for her contribution to the reviewing process, and QatarDebate Center for providing funding and resources for the shared task.

References

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. ARBERT & MARBERT: Deep bidirectional transformers for Arabic. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7088–7105, Online. Association for Computational Linguistics.

Abdelrahman Ahmad Saeed Abdelaziz, Hend Emad, and Ensaf H. Mohamed. 2026. NU_Analytics at Daleel 2026: Independent label scoring and structured span prediction for Arabic argumentative discourse mining. Budapest, Hungary. Association for Computational Linguistics.

Khalid Al-Khatib, Henning Wachsmuth, Matthias Hagen, and Benno Stein. 2017. Patterns of argumentation strategies across topics. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 1351–1357, Copenhagen, Denmark. Association for Computational Linguistics.

Khalid Al-Khatib, Henning Wachsmuth, Johannes Kiesel, Matthias Hagen, and Benno Stein. 2016. A news editorial corpus for mining argumentation strategies. In *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, pages 3433–3443, Osaka, Japan. The COLING 2016 Organizing Committee.

Rabee Al-Qasem. 2026. RabeeQasem - ImpactAi at Daleel 2026: Conformer-based Arabic argumentative discourse mining. Budapest, Hungary. Association for Computational Linguistics.

Abdul Gabbar Al-Sharafi, Mohammad Majed Khader, Mohamed Ahmed, Mohamad Hamza Al-Sioufy, Wajdi Zaghouani, and Ali Al-Zawqari. 2025. A hybrid annotation model for Arabic argumentative debate corpus. In *Arabic Language Processing: From Theory to Practice*, volume 2339 of *Communications in Computer and Information Science*, pages 97–113. Springer, Cham.

Ali Al-Zawqari, Mohamed Ahmed, Abdul Gabbar Al-Sharafi, Mohammad M. Khader, Ali Safa, and Gerd Vandersteen. 2025. Neural classification of argument elements and styles in Arabic competitive debates. *IEEE Access*, 13:115944–115959.

Firoj Alam, Hamdy Mubarak, Wajdi Zaghouani, Giovanni Da San Martino, and Preslav Nakov. 2022. Overview of the WANLP 2022 shared task on propaganda detection in Arabic. In *Proceedings of the Seventh Arabic Natural Language Processing Workshop (WANLP)*, pages 108–118, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.

Salah Ali Abdo. 2026. SalahAbdo at Daleel 2026: Routing each label to LLM votes, a fine-tuned encoder, or sequence structure. Budapest, Hungary. Association for Computational Linguistics.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2020. AraBERT: Transformer-based model for Arabic language understanding. In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection*, pages 9–15, Marseille, France. European Language Resource Association.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2021. AraELECTRA: Pre-training text discriminators for Arabic language understanding. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 191–195, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.

Dimitar Dimitrov, Bishr Bin Ali, Shaden Shaar, Firoj Alam, Fabrizio Silvestri, Hamed Firooz, Preslav Nakov, and Giovanni Da San Martino. 2021. SemEval-2021 task 6: Detection of persuasion techniques in texts and images. In *Proceedings of the 15th International Workshop on Semantic Evaluation (SemEval-2021)*, pages 70–98, Online. Association for Computational Linguistics.

Khalid N. Elmadani, Bashar Alhafni, Hanada Taha, and Nizar Habash. 2025. BAREC shared task 2025 on Arabic readability assessment. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 239–252, Suzhou, China. Association for Computational Linguistics.

Ivan Habernal and Iryna Gurevych. 2015. Exploiting debate portals for semi-supervised argumentation mining in user-generated web discourse. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 2127–2137, Lisbon, Portugal. Association for Computational Linguistics.

Ivan Habernal and Iryna Gurevych. 2017. Argumentation mining in user-generated web discourse. *Computational Linguistics*, 43(1):125–179.

Maram Hasanain, Firoj Alam, Hamdy Mubarak, Samir Abdaljalil, Wajdi Zaghouani, Preslav Nakov, Giovanni Da San Martino, and Abed Freihat. 2023. ArAIEval shared task: Persuasion techniques and disinformation detection in Arabic text. In *Proceedings of ArabicNLP 2023*, pages 483–493, Singapore (Hybrid). Association for Computational Linguistics.

Maram Hasanain, Md. Arid Hasan, Fatema Ahmad, Reem Suwaileh, Md. Rafiul Biswas, Wajdi Zaghouani, and Firoj Alam. 2024. ArAIEval shared task: Propagandistic techniques detection in unimodal and multimodal Arabic content. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 456–466, Bangkok, Thailand. Association for Computational Linguistics.

Annette Hautli-Janisz, Zlata Kikteva, Wassiliki Siskou, Kamila Gorska, Ray Becker, and Chris Reed. 2022. QT30: A corpus of argument and conflict in broadcast debate. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 3291–3300, Marseille, France. European Language Resources Association.

Christopher Hidey, Elena Musi, Alyssa Hwang, Smaranda Muresan, and Kathy McKeown. 2017. Analyzing the semantic types of claims and premises in an online persuasive forum. In *Proceedings of the 4th Workshop on Argument Mining*, pages 11–21, Copenhagen, Denmark. Association for Computational Linguistics.

Turab Hussain Usmani. 2026. Solodiscourse at Daleel 2026: Lexically-augmented focal-loss ensembling for Arabic argumentative discourse unit classification. Budapest, Hungary. Association for Computational Linguistics.

Go Inoue, Bashar Alhafni, Nurpeiis Baimukan, Houda Bouamor, and Nizar Habash. 2021. The interplay of variant, size, and task type in Arabic pre-trained language models. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 92–104, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.

Mohammad M. Khader, AbdulGabbar Al-Sharafi, Mohamad Hamza Al-Sioufy, Wajdi Zaghouani, and Ali Al-Zawqari. 2024. Munazarat 1.0: A corpus of Arabic competitive debates. In *Proceedings of the 6th Workshop on Open-Source Arabic Corpora and Processing Tools (OSACT) with Shared Tasks on Arabic LLMs Hallucination and Dialect to MSA Machine Translation*, pages 20–30, Torino, Italia. ELRA and ICCL.

John Lawrence and Chris Reed. 2019. Argument mining: A survey. *Computational Linguistics*, 45(4):765–818.

Yann Mathet, Antoine Widlöcher, and Jean-Philippe Métivier. 2015. The unified and holistic method gamma for inter-annotator agreement measure and alignment. *Computational Linguistics*, 41(3):437–479.

Mohammed Mohaisen. 2026. BME-Daleel at Daleel 2026: Joint modeling, structured decoding, and domain-routed ensembles for Arabic argument units. Budapest, Hungary. Association for Computational Linguistics.

Sara Nabhani and Khalid Al Khatib. 2025. Transfer or translate? argument mining in Arabic with no native annotations. In *Proceedings of The Third Arabic Natural Language Processing Conference*, pages 407–416, Suzhou, China. Association for Computational Linguistics.

Sara Nabhani, Claudia Borg, Kurt Micallef, and Khalid Al-Khatib. 2025. Integrating argumentation features for enhanced propaganda detection in Arabic narratives on the Israeli war on Gaza. In *Proceedings of the 1st International Workshop on Nakba Narratives as Language Resources*, pages 127–149, Abu Dhabi, UAE. Association for Computational Linguistics.

Ahmed Osrf, Eid Abo Baid, and Shadi Abudalfa. 2026. BURHAN at Daleel shared task subtask-1: A multi-label transformer pipeline for Arabic argumentative discourse unit classification. Budapest, Hungary. Association for Computational Linguistics.

Ramon Ruiz-Dolz, John Lawrence, Ella Schad, and Chris Reed. 2024. Overview of DialAM-2024: Argument mining in natural language dialogues. In *Proceedings of the 11th Workshop on Argument Mining (ArgMining 2024)*, pages 83–92, Bangkok, Thailand. Association for Computational Linguistics.

Amal Saad Alshehri, Nelly Bencomo, Housam Babiker, and Amir Atapour-Abarghouei. 2026. Durham-Jazan at Daleel 2026: Domain-routed ensembles, LLM augmentation, and the limits of decision-rule tuning for Arabic argument mining. Budapest, Hungary. Association for Computational Linguistics.

Karla Salas-Jimenez, Jean-Guy Mailly, and Leila MOUDJARI. 2026. AIDAL-ARG at Daleel 2026: Hybrid and multitask approaches for Arabic argument mining. Budapest, Hungary. Association for Computational Linguistics.

Eman Sedqy Shlkamy and Wajdi Zaghouani. 2026. Northwestern 12 at Daleel 2026: CAMeLBERT-Mix for multi-label Arabic argumentative discourse unit classification. Budapest, Hungary. Association for Computational Linguistics.

Noam Slonim, Yonatan Bilu, Carlos Alzate, Roy Bar-Haim, et al. 2021. An autonomous debating system. *Nature*, 591(7850):379–384.

Christian Stab and Iryna Gurevych. 2014. Identifying argumentative discourse structures in persuasive essays. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 46–56, Doha, Qatar. Association for Computational Linguistics.

Christian Stab and Iryna Gurevych. 2017. Parsing argumentation structures in persuasive essays. *Computational Linguistics*, 43(3):619–659.

Bhuvanesh Verma, Ali Abusaleh, and Alexander Mehler. 2026. TTLab at Daleel 2026: STAR-Ar sequence tagging for argument recognition in Arabic. Budapest, Hungary. Association for Computational Linguistics.

Wajdi Zaghouani, Abdelhamid Ahmed, Xiao Zhang, and Lameya Rezk. 2024. QCAW 1.0: Building a Qatari corpus of student argumentative writing. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 13382–13394, Torino, Italia. ELRA and ICCL.